

Surveillance video synopsis generation method via keeping important relationship among objects

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Yumin Tian¹, Haihong Zheng¹, Qichao Chen¹ ✉, Dan Wang¹, Risan Lin¹¹School of Computer Science and Technology, Xidian University, Xi'an, Shannxi 710071, People's Republic of China

✉ E-mail: qichaochen2016@163.com

Abstract: To reduce the efforts of human in browsing long surveillance videos, synopsis videos are proposed. Traditional synopsis video generation methods condense most of the activities in the video by simultaneously showing several actions, even when they originally occurred at different times. This inevitably causes ignorance of temporal relationship among objects. For example, two persons walk shoulder to shoulder and they are detected and tracked separately, but in the synopsis they never 'met'. In this study, a trajectory mapping model is defined, whose energy function includes not only the cost caused by the synopsis video, but that of the original video. In this way, it tries to make the relationship between objects of the original video consistent with that of the synopsis. Finally, the video synopsis is generated by an energy minimisation method. Experiments show that the proposed video synopsis can reduce the spatiotemporal redundancies of the input video as much as possible. Moreover, it can keep the important relationship between objects and maintain the time consistency of important activities.

1 Introduction

With the development of surveillance cameras, recording daily events which happen in the environments becomes possible. However, how to efficiently browse such a huge number of videos becomes a challenging problem in visual surveillance. To provide a fast browsing mode, some surveillance systems [1] record key frames of foreground objects in the scene. However, these static key frames do not contain continuous object motions, which are important for event analysis. Thus, video abstraction methods are proposed recently, in which significant images or video segments are retrieved to represent the original videos.

Object-based video synopsis [2–9] analyses the structure and content of the video, extracts meaningful objects from the original video, and somehow combines them to form a video summary to sufficiently express the semantic content of the video. Video synopsis techniques can make several hours of video condensed into 10 min or less, significantly reducing the browsing time of the video, and the basic structure of the original video taking 'frame' as a unit will be changed to take 'object' as the basic structure of the unit, which can make viewers find the interesting target quickly

1.1 Related work on video synopsis

Many approaches have been proposed for condensing the volume of videos. Hong-Wen *et al.* [2] extracted informative space–time video portions that can be moved and stitched using first-fit and graph-cut optimisations which maximises the amount of visual information. As optimal boundaries between portions are found, visually unpleasant seams usually appear between portions that do not match. Rav-Acha *et al.* [7] proposed an object-based approach for video synopsis that is similar to that of Hong-Wen *et al.* [2]. Both of these methods change the chronological order of objects, and show several actions at the same time. The difference is that the latter only moves objects along the time axis. Pritch *et al.* [8] extended the later work to process always-on videos captured by surveillance cameras or webcams. The more complete work is presented in [9], in which they first extracted interesting objects (tubes), and then moved them along the time axis while preserving activities and local chronological orderings. It can handle always-on videos and take illumination-varying backgrounds into account. However, since it processes moving objects only in the temporal domain, this method cannot fully utilise the video space in the spatial domain and may produce collision artefacts when the

synopsis is compressed much or contains more moving objects. Yongwei *et al.* [10] globally shifts moving objects in both spatial and temporal domains, which shifts objects temporally to reduce the length of the video and moves colliding objects spatially to avoid visible collision artefacts. Since these synopsis video generation methods applying optimisation on video tubes are very time-consuming and infeasible for real-time online generation, the feasibility of synopsis video generation in practical situations is significantly reduced. To solve this problem, many on-line synopsis generation methods [11, 12] have been proposed. Chun-Rong *et al.* [12] formulate the synopsis video generation problem as a maximum posteriori probability estimation problem, where the positions and appearing frames of video objects are chronologically rearranged in real-time without needing to know their complete trajectories. Moreover, a synopsis table is employed with maximum posteriori probability estimation to decide the temporal locations of the incoming foreground objects in the synopsis video without an optimisation procedure.

1.2 Object-based video synopsis

The video synopsis suggested in this paper is similar to [7]. Objects of interest are defined and viewed as tubes in the space–time volume. A temporal shift is applied to each object, creating a shorter video synopsis while avoiding collisions between objects [13]. The key difference is shown in the following three properties: (i) A long-time moving object is not broken into segments to improve the readability. (ii) To reflect all the circumstances of the original video veritably, and not lose any event information, all interest activities should appear in the synopsis, we defined activity energy cost which is the penalty function for an object tube that does not appear in synopsis. (iii) We defined four object relationships between original video and synopsis: the relationship between positive sequence, the relationship between reverse sequence, the relationship between true-collision, and the relationship between false-collision. Synopsis video should keep the important temporal relationships among different objects in original video as much as possible. For example, object A following object B appears, and the two objects are chasing each other in the scene. This is a kind of relationship between two objects, which should be retained in the video synopsis. The costs of the related positive sequence and reverse order are the penalty function brought by breaking the hidden dependency. Keeping the

video synopsis shorter than the original while avoiding conflict and occlusion between trajectories meanwhile is a challenge. There are two kinds of collisions in the trajectory combination process. One is real-collision, which targets colliding with each other in the original video. For example, two people are walking on the road and talking to each other with rubbing shoulders. The other is false-collision that targets do not collide with each other in the original video but collide in the synopsis. If only false-collision problem is considered, the relevant information of the real-collision will be lost, and the true-collision objects may be not included in the synopsis. This may cause the length of the synopsis being longer than that of the original. The last property is the main contribution of our method.

This paper is organised as follows. Section 2 describes the trajectory mapping model (TMM) in detail. To solve the optimisation problem, Section 3 gives an energy minimisation method. Experiments are shown in Section 4. Finally, conclusions are drawn in Section 5.

2 Trajectory mapping model

In this section, the input video is supposed to have extracted the background with Gaussian mixture model [14], and object trajectory is got by using tracking algorithm mentioned in [15]. Then, the TMM is described in detail.

Let N frames of an input video sequence I be represented in a space-time volume $I(x, y, t)$, where (x, y) is the spatial coordinate of a pixel in frame t , satisfying $(1 \leq x \leq W)$, $(1 \leq y \leq H)$, and $(1 \leq t \leq N)$. Let $b_m = \{B_t^m | t \in t_m\}$, $t_m = [t_m^s, t_m^e]$ be the trajectory of object m , which is the set of all bounding rectangles B_t^m for the object from entering to leaving in the scene. t_m represents the period object appearing. So, the feature of trajectory b_m can be defined as follows:

$$x_m(x, y, t) = \begin{cases} 1 & x, y \in B_t^m, t \in t_m \\ 0 & \text{else} \end{cases} \quad (1)$$

where $x_m(x, y, t)$ represents blob that detected is in trajectory B_t^m or not. Therefore, object trajectories $BO = \{b_i | 1 \leq i \leq n\}$ of the original video are formed by all the object trajectories. n is the number of the objects. The idea of video synopsis generation is to rearrange the object trajectory along the time, so the length of synopsis S is much shorter than that of the input video. For trajectory b_m , if b'_m is the corresponding trajectory in the synopsis, whose appearing period is $t'_m = [t_m'^s, t_m'^e]$, and BO' is the object trajectory set of the synopsis, then the synopsis is generated with a mapping model M that satisfies

$$M(BO) = BO' \quad (2)$$

To keep the temporal relationships between objects of original videos retained, an energy function $E(M)$ is defined as follows:

$$E(M) = E_a(BO) + \alpha E_{tps}(BO) + \beta E_{tns}(BO) + \gamma E_{tc}(BO) + \lambda E_{tct}(BO) \quad (3)$$

where $E_a(BO)$ is the activity energy cost, $E_{tps}(BO)$ is the cost of the related positive sequence between objects, $E_{tns}(BO)$ is the cost of the related reverse sequence between objects, $E_{tc}(BO)$ is the cost of the false-collision, $E_{tct}(BO)$ is the cost of real-collision, and α, β, γ , and λ are the weight coefficients.

As $E(M)$ is introduced above, the best solution of the mapping can be given as follows:

$$E(M_{opt}) = \text{Min}_{M \in \text{MAP}} (E(M)) \quad (4)$$

where M is all possible mapping solutions.

2.1 Activity energy cost

Generally, moving objects with large size and long trajectory are the interest object, while the object with smaller area and transient occurrence time may be unimportant information or noise due to detection result, so the activity energy cost of object b_i is given as follows:

$$E_a(b_i) = \sum_{t \in t_m} \left(\frac{\text{Area}(b_t)}{\text{Area}(f_t)} \right) \quad (5)$$

where f_t is a video frame.

2.2 Costs of the related positive sequence and reverse order

The penalty function of the related positive sequence $E_{tps}(b_m)$ for object trajectory b_m is shown in (6), where np is the number of the object trajectories that appears ahead of b_m in the synopsis while it should be at the back of b_m in the original video, K is an experimental factor, and i_m is the tracking index of object b_m in original video

$$E_{tps}(b_m) = e^{K * np - i_m} \quad (6)$$

The penalty function of the related reverse order $E_{tns}(b_m)$ for object trajectory b_m of the current frame is updated based on previous frame, shown in (7), where $E_{tn}(b_m, b_b)$ is the generated reverse cost for b_m , if b_b is added first in the synopsis, shown in (8), t_b^s and t_m^s are the trajectory start time of b_b and b_m in the original video

$$E_{tns}(b_m) = E_{tns}(b_m) + E_{tn}(b_m, b_b) \quad (7)$$

$$E_{tn}(b_m, b_b) = e^{i_m - i_b} * (t_b^s - t_m^s + 1) \quad (8)$$

2.3 Collision energy cost

To keep the real relationship between objects, the penalty function of false-collision $E_{tc}(b_m, b_b)$ for the object trajectories b_m and b_b is defined as follows:

$$E_{tc}(b_m, b_b) = - \sum_{x, y, t \in t_b' \cap t_m'} x_m(x, y, t') * x_b(x, y, t') \quad (9)$$

where t' is the current frame in the synopsis, t_m' and t_b' are the time period in the synopsis for object trajectories b_m and b_b , respectively.

Energy cost of real-collision $E_{tct}(b_m, b_b)$ is included to keep real-collision information

$$E_{tct}(b_m, b_b) = \sum_{x, y, t \in t_b \cap t_m} x_m(x, y, t) * x_b(x, y, t) \quad (10)$$

where t is the current frame in the original video, t_m and t_b are the time period in the original video for object trajectories b_m and b_b , respectively.

3 TMM with energy minimisation method

To find the optimum of trajectory mapping, the energy function given by (4) can be expressed as follows:

$$E(M) = \sum_{m \in BO} E_a(b_m) + \alpha \sum_{m \in BO} E_{tps}(b_m) + \beta \sum_{m \in BO} E_{tns}(b_m) + \gamma \sum_{m, b \in BO} E_{tc}(b_m, b_b) + \lambda \sum_{m, b \in BO} E_{tct}(b_m, b_b) \quad (11)$$

The optimal solution of (11) can be obtained by genetic algorithm. Genetic algorithm depends on the traversal times of the genetic operators. Suppose there are M object trajectories and the synopsis

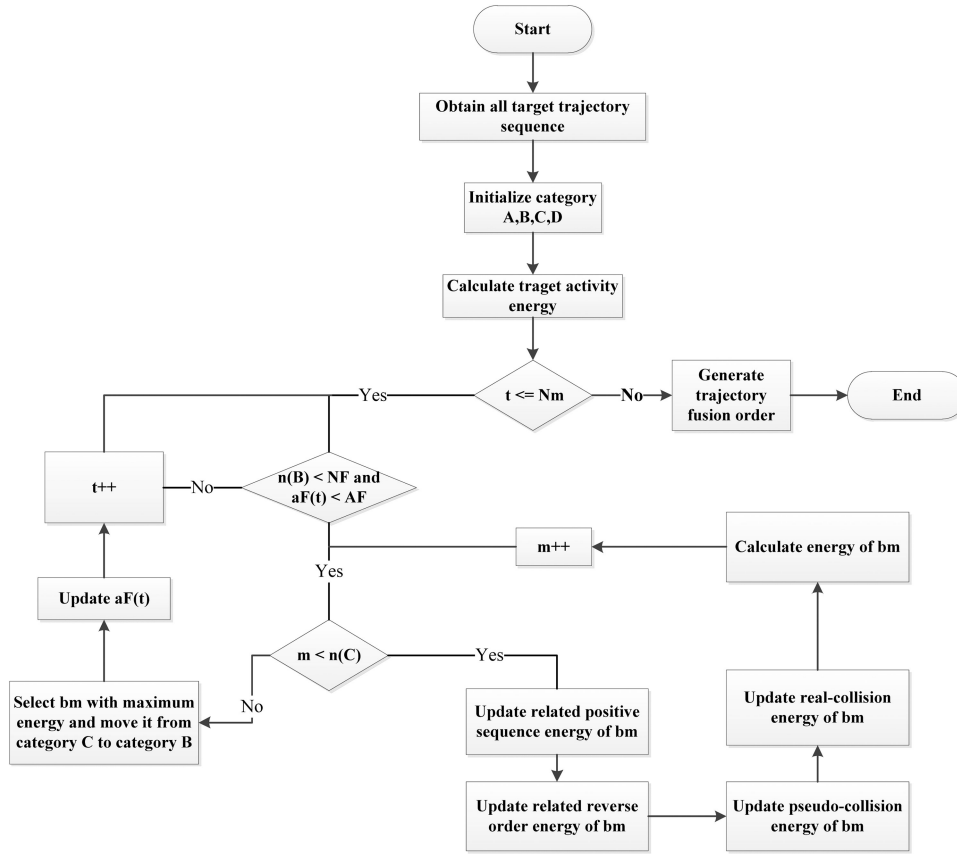


Fig. 1 Algorithm flow diagram of the video synopsis generation

video have N frames, and then in extreme cases it needs C_N^M time of operations to reach the optimal solution by traversing all the solutions. Moreover, if trajectory objects' information cannot be represented completely, different results will be achieved when run several times.

To solve the above problem, a solution of the optimal mapping by simplifying (11) is implemented. First, the object of BO in the current frame is classified into four categories:

C_A : $b_m \in BO'$, and the trajectory tube has been added completely in the current frame.

C_B : $b_m \in BO'$, but the trajectory tube has not been fully added in the current frame, the remaining part will be added in the subsequent frame.

C_C : $b_m \in BO$ and $b_m \notin BO'$, the object trajectory tube has not been added yet.

C_D : $b_m \in BO$ and $b_m \in C_C \cap C_B$, that is the object trajectory happening to truly collide with the object being added in the current frame.

For each frame $S(x, y, t')$ of the synopsis, when the current frame is not filled, the energy of mapping function is $E(M, t')$ and it can be written as follows: (see (12))

$$E_{tps}(b_{m'}, t') = e^{K * |C_A \cup C_B| - i_m} \quad (13)$$

$$E_{tns}(b_{m'}, t') = \sum_{m \in C_C, m \in C_A \cup C_B} E_{tn}(b_{m'}, b_b) \quad (14)$$

where $C_A \cup C_B$ is the number of objects in the union set of categories C_A and C_B , K is the same with which in (6), and $E_{tn}(b_m, b_b)$ in (14) is the same as that in (8).

The optimum solution for the mapping function is to make the energy function minimisation. That is, for each frame of synopsis, objects that with larger activity energy in category C_C , larger accumulated energy of positive sequence, and reverse order is added first to avoid further accumulation, smaller energy of the false-collision in category C_B and a larger energy of real-collision in category C_D .

Suppose $n(B)$ is the number of the objects being filled in each synopsis frame, N_F is the number that is allowed accommodating per frame. Moreover, $a_F(t)$ is the screen filled degree for each frame that is defined

$$a_F(t) = \bigcup_{m \in C_B} x_m(x, y, t) \quad (15)$$

When the number of the objects $n(B)$ is less than or equal to N_F and the screen filled degree $a_F(t)$ is less than or equal to the maximum screen filling degree A_F , the best object b_m from category C_C is selected for the synopsis frame to fill according to (12), and b_m is moved from categories C_C to C_B , and the energy information of the objects are updated in category C_C meanwhile. Repeating the process until the condition is not satisfied and then it starts to fill next synopsis frame. The process is shown in Fig. 1.

If M object trajectories are reassembled to a video synopsis with N frames, the algorithm's complexity of trajectory combination is $O(N * M^2)$, $M < N$.

$$E(M, t') = \sum_{m \in C_C} E_a(b_m) + \alpha \sum_{m \in C_C} E_{tps}(b_{m'}, t') + \beta \sum_{m \in C_C} E_{tns}(b_{m'}, t') + \gamma \sum_{m \in C_C, b \in C_B} E_{tc}(b_{m'}, b_b) + \lambda \sum_{m \in C_C, b \in C_D} E_{tct}(b_{m'}, b_b) \quad (12)$$

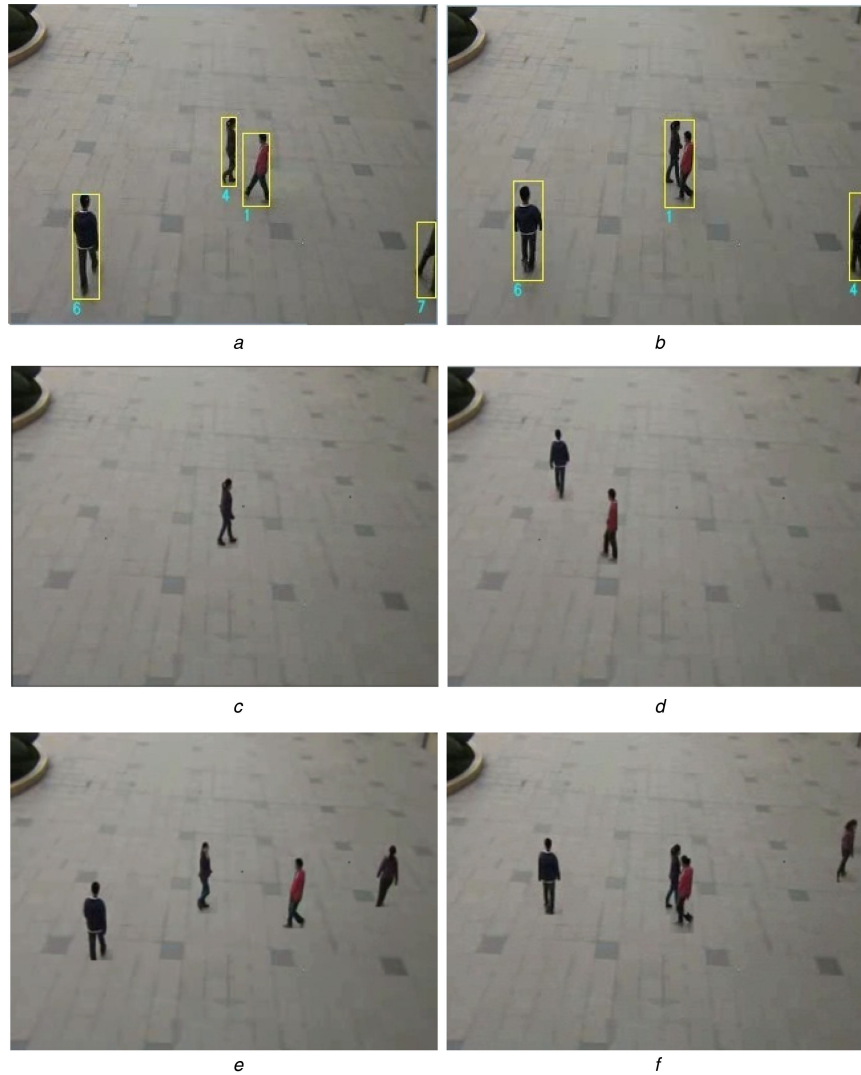


Fig. 2 Experimental results of control false-collision and true-collision energy

(a) Thirty-ninth frame in original video, (b) Fifty-sixth frame in original video, (c) Forty-first frame in synopsis without considering true-collision, (d) One hundred fiftieth frame in synopsis without considering true-collision, (e) Twentieth frame in synopsis considering true-collision, (f) Fifty-fifth frame in synopsis considering true-collision

4 Experimental results

4.1 Experiments of important relationships between objects

As shown in Fig. 2 the length of the testing video is 139 frames, in which objects 1 and 4 occur to interlace with each other in the original video Figs. 2a and b, and the occurrence time of the two objects are both 120 frames.

In extreme case, suppose false-relationship is forbidden totally while the energy cost of true-collision is not considered, the synopsis is shown in Figs. 2c and d. In this case, the length of the synopsis is 239 frames. Objects 1 and 4 are separated which is not consistent with the original information. Moreover, it causes the length of the synopsis longer than that of the original video.

In above experiment, if considering the energy cost of true-collision in trajectory combination, the length of the synopsis is 124 frames as shown in Figs. 2e and f. It is obvious that the behaviour of objects 1 and 4 is similar with the original video. Moreover, the length of the synopsis is shorter than the original video.

4.2 Experiments of video synopsis generation

For quantitative comparisons, two performance metrics are used and compared it with our implementation of the method in [13, 16]. The first metric is the frame reduction (FR) rate, which is computed as the ratio of the number of the frames of the synopsis to the number of the frames of the original video. The smaller the FR, the fewer frames the synopsis contains. The second one is the

object loss rate (OLR), which is computed as the ratio of the number of loss objects in the synopsis video to that in the original video.

For expression convenience, we also involve the following parameters. (i) M_B is the number of objects. (ii) N_F is the maximum target number that can be accommodated in each synopsis frame.

Table 1 shows the synopsis on condition that OLR is 0. Table 2 shows the result on condition that FR is 10. Table 1 shows that with N_F increases, the FR decreases. In Table 2, with $N_F = 10$, the FR decreases further.

The experimental results compared with [13, 16] which we can see the FR rate is better, and our method can get higher FR by directly setting the parameters while keeping objects as much as possible. Our method keeps object trajectories' original relationship and avoid collision while traditional method [7] loses relationship when the FR is high.

Fig. 3 shows example of two frames in the synopsis. Fig. 3a is the result of $N_F = 10$, the generated synopsis frame has a high screen filling degree and the resulting video summary is short. Fig. 3b shows that when $N_F = 3$, the targets in the synopsis frame are sparse and spatial information is not effectively utilised. Accordingly the generated synopsis video is longer.

5 Conclusion

A synopsis generation algorithm based on TMM with classified trajectory into four categories is proposed, which can remove the space-time redundant information of the surveillance video. Since

Table 1 Comparison result of FR with OLR = 0

No. of original video	M_B	N_F	No. of synopsis video	FR	OLR
1260	19	10	525	0.41	0
1260	19	3	693	0.55	0
2237	60	10	551	0.24	0
2237	60	3	998	0.44	0
4320	20	10	1050	0.24	0
4320	20	3	1370	0.31	0
15345	98	10	2572	0.17	0
15345	98	3	3711	0.24	0

Table 2 Comparison result of FR and OLR with $N_F = 10$

No. of original video	M_B	N_F	No. of synopsis video	FR	OLR
1260	19	10	100	0.10	0.44
1260	19	10	200	0.20	0.25
2237	60	10	200	0.09	0.33
2237	60	10	130	0.06	0.48
4320	20	10	275	0.06	0.23
4320	20	10	550	0.12	0.36
15345	98	10	1484	0.1	0.35
15345	98	10	747	0.05	0.52

**Fig. 3** Comparison of synopsis video frames with different N_F

(a) Twenty-ninth frame in the synopsis with $N_F = 10$, (b) Twenty-seventh frame in the synopsis with $N_F = 3$

the cost function defined cost function depends not only on synopsis video but also on original video, it can overcome the problem of losing important relationship between objects in traditional algorithms. With synopsis generation algorithm via keeping important relationship among objects, it can generate a short video that will reserve the invisible correlativity information, and meet the needs of quick browsing for users.

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